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Identifying a non-chiral ligand for the efficient chirality transfer in carbonylation of propargylic mesylates†

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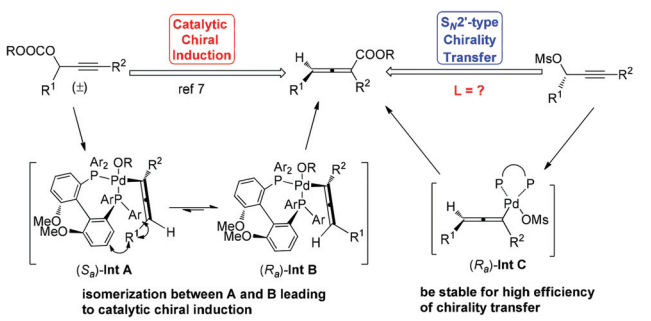
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Enantioselective synthesis of optically active 2,4-disubstituted 2,3-allenoates has been realized with a very high chirality transfer efficiency using the inexpensive commercially available non-chiral DPEphos in the presence of 100–200 psi of CO at room temperature. The key for the high efficiency of this chirality transfer process has been certified as the matched coordination of Pd with DPEphos forming a relatively stable optically active square planar allenylic Pd(II) complex, which undergoes a facile carbonylation in the presence of 10 equiv. of BnOH and 100–200 psi of CO.

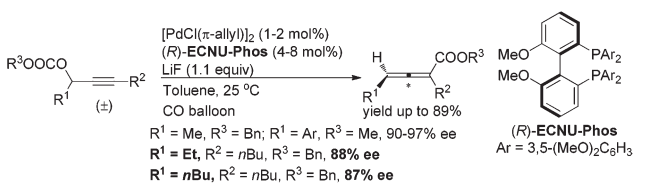
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Allenes have been recognized as an important type of starting materials for organic synthesis;[1] thus, asymmetric synthesis of allenes, especially 2,3-allenoates, is of high current interest in organic chemistry.[2–8] Recently, it was reported that optically active 2,3-allenoates have been prepared *via* Pd-catalyzed carbonylation *via* catalytic chiral induction from the corresponding racemic starting materials, in which the racemization between (S_a)- and (R_a)-allenylic palladium complexes **Int A** and **Int B** is found to be crucial for the catalytic chiral induction strategy (Scheme 1).[8] On the other hand, we have been pursuing a non-chiral ligand for a high efficiency of chirality transfer, which may importantly supplement the above mentioned reaction for R^1 being an alkyl group beyond methyl (eqn (1))[8] based on the following concept: a ligand for matched coordination with Pd to prevent interconversion between (R_a)-**Int C** and (S_a)-**Int C** (Scheme 1) followed by a very facile carbonylation.[9]

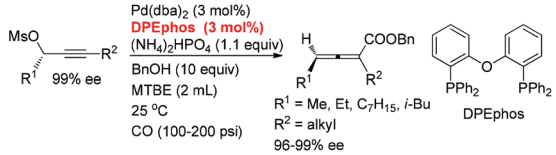


Scheme 1 Concepts for the enantioselective synthesis of 2,3-allenoates *via* Pd-catalyzed carbonylation.

Here, we wish to report the successful identification of such a non-chiral cheap and readily available bidentate DPEphos as the ligand for the Pd-catalyzed efficient chirality transfer of enantio-enriched non-terminal propargylic mesylates forming optically active 2,3-allenoates (eqn (2)).[7]



(1)



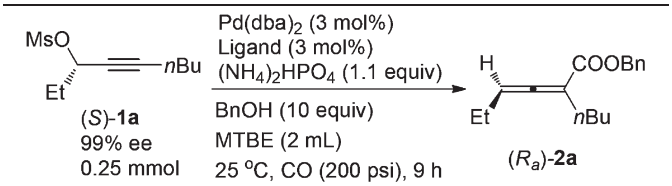
(2)

A screening of the non-chiral ligand has been carried out for the high efficiency of chirality transfer of (*S*)-**1a**:[7e] electron-rich tri(2,4,6-trimethoxyphenyl)phosphine (**L1**) was used for the reason of a better coordination ability with Pd to prevent racemization of (R_a)-**Int C** (Scheme 1); however, (R_a)-**2a** was

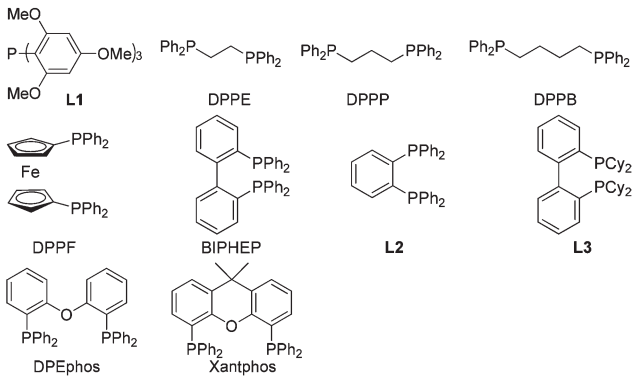
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Table 1 Ligand effect in Pd-catalyzed carboxylation of enantioenriched propargylic mesylate (*S*)-**1a**



Entry	Ligand	Bite angle[a]	Yield%[b] of (R_a)-**2a**[c] (ee%)
1	**L1**	—	26 (36)
2	DPPE	86[11a]	66 (24)
3	DPPP	91[11a]	73 (40)
4	DPPB[d]	94[11a]	75 (8)
5	DPPF	99[11a]	66 (89)
6	BIPHEP	92[11b]	77 (93)
7	**L2**[e]	87[11a]	85 (24)
8	**L3**[f]	92	79 (98)
9	DPEphos[d]	104[11a]	82 (97)
10	Xantphos[d]	108[11a]	70 (76)

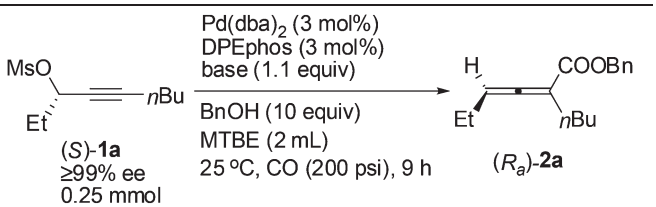


[a] Bite angles were obtained from crystal structures of bis(phosphine)$PdCl_2$ complexes. [b] Isolated yield. [c] ee value of the product was determined by HPLC. [d] The bite angle was obtained from a crystal structure of the bis(phosphine)Pd(1,1-dimethylallyl)tetrafluoroborate complex. [e] The bite angle was obtained from a crystal structure of the bis(phosphine)$PdBr_2$ complex. [f] The bite angle was estimated based on analogy to BIPHEP.

obtained in only 26% yield and 36% ee (Table 1, entry 1). Then we started to work with bidentate ligands for the same purpose: the chirality transfer efficiency was rather low with DPPE, DPPP or DPPB as the ligand (Table 1, entries 2–4). Surprisingly, DPPF performed much better and the enantiopurity of the product reached 89% (Table 1, entry 5)! The ee value of the product further reached 93% with BIPHEP (Table 1, entry 6). Thus, we started to correlate the bite angles of some bidentate ligands with the ee of the products.[10] 1,2-Bis(diphenylphosphino)benzene (**L2**) led to a poor chirality transfer efficiency (Table 1, entry 7), and it was consistent with the result of using DPPE (Table 2, comparing entry 2 with entry 7), as the bite angles of **L2** and DPPE are almost the same. Interestingly, replacing the phenyl group in BIPHEP with a more electron-donating cyclohexyl group (**L3**) led to the best result: the chirality transfer efficiency reached almost 100%, which must have been caused by the better electron-donating ability of this PCy_2 group (Table 1, entry 8). With an increase of the bite angle, the chirality transfer efficiency was higher: when DPEphos with a bite angle of 104° was used, the optically active 2,3-allenoate (R_a)-**2a** was also afforded with 97% ee (Table 1, entry 9).[11] However, when the bite angle became even bigger, the enantioselectivity dropped sharply: (R_a)-**2a** was only obtained in 70% yield with 76% ee using Xantphos with a bite angle of 108°, only 4° greater than DPEphos as the ligand (Table 1, entry 10). As the price of DPEphos is much lower than **L3**, it has been finally considered as the ligand of choice for further study.[12]

With DPEphos as the ligand, a series of bases were then examined (Table 2): using either $(NH_4)_2HPO_4$ or $(NH_4)_2CO_3$ led to the same chirality transfer efficiency with a slight difference of the isolated yield (Table 2, entries 1 and 2); however, the enantiopurity of (R_a)-**2a** dropped to 92% when K^+ took the place of NH_4^+ (Table 2, comparing entry 2 with entry 3); (R_a)-**2a** could be obtained with 96% or 95% ee when $NaHCO_3$ or KH_2PO_4 was used (Table 2, entries 4 and 5); interestingly, LiF, which performed best in the **ECNU-Phos**-induced reaction,[7] led to a much lower yield (64%) albeit in 97% ee (Table 2, entry 6); the performance of organic bases was poor as compared to inorganic ones: (R_a)-**2a** was obtained in 75% yield with approximately 90% ee when pyridine or TEA was used (Table 2, entries 7 and 8).

Table 2 Base effect in DPEphos-assisted carboxylation of enantioenriched (*S*)-**1a**



Entry	Base	Yield% of (R_a)-**2a**[a]	ee% of (R_a)-**2a**[b]
1	$(NH_4)_2HPO_4$	82	97
2	$(NH_4)_2CO_3$	80	97
3	K_2CO_3	75	92
4	$NaHCO_3$	79	96
5	KH_2PO_4	73	95
6	LiF	64	97
7	Py	75	91
8	TEA	75	90

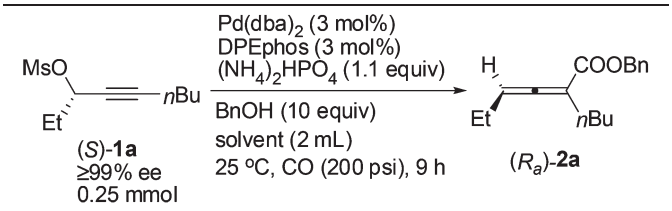
[a] Isolated yield. [b] ee value of the product was determined by HPLC.

A screening of the solvent was subsequently conducted. When ether, such as MTBE, 1,4-dioxane and THF, or toluene was used, the enantiopurity of (R_a)-**2a** was almost the same (96–97%), and the reaction in THF led to only 69% yield of (R_a)-**2a** (Table 3, entries 1–4). The enantiopurity of (R_a)-**2a** dropped to 87% with DMF as the solvent (Table 3, entry 5); the

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Table 3 Solvent effect in DPEphos-assisted carboxylation of enantio-enriched (*S*)-**1a**[a]



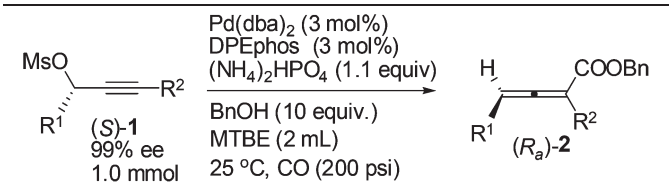
Entry	Solvent	Yield% of (R_a)-**2a**[b]	ee% of (R_a)-**2a**[c]
1	MTBE	82	97
2	1,4-Dioxane	78	96
3	THF	69	96
4	Toluene	76	97
5	DMF	69	87
6	CH_3CN	77	5

[a] MTBE = methyl *tert*-butyl ether, DMF = *N*,*N*′-dimethylformamide, THF = tetrahydrofuran. [b] Isolated yield. [c] ee value of the product was determined by HPLC.

reaction proceeded in CH_3CN with a very poor chirality transfer efficiency (Table 3, entry 6).

The scope of this reaction was then examined with 1 mmol of enantio-enriched non-terminal propargylic mesylates at room temperature and 200 psi of CO with 3 mol% each of $Pd(dba)_2$ and DPEphos as the catalyst (Table 4). When the R^1 group is ethyl, the enantioselectivities of the substrates with different alkyl groups were all greatly improved as compared with our previous report with (*S*)-Segphos as the ligand (Table 4, entries 1 and 3–7).[7f] And the chlorine substituted substrate (*S*)-**1g** led to the corresponding product with 97% ee (Table 4, entry 8). Furthermore, when the unsaturated groups, such as phenyl or vinyl, were introduced into R^1, the corresponding products were also obtained with 98% ee, an obvious improvement again as compared to the (*S*)-Segphos-assisted protocol[7f] (Table 4, entries 9 and 10). Changing the chain length of R^1 did not influence the outcome: the methyl-substituted 2,3-allenoate (R_a)-**2j** was obtained in 80% yield with 98% ee (Table 4, entry 11); and 2,3-allenoate (R_a)-**2k** with a longer *n*-heptyl group was also afforded in 84% yield with 96% ee (Table 4, entry 13); even the bulky groups performed very well in this process: with R^1 being i-butyl, the corresponding (R_a)-**2l** was prepared in 85% yield with 97% ee (Table 4, entry 15). Furthermore, 4-alkyl substituted 2,3-allenoates, which were obtained with less than 90% ee unless R^1 is methyl in (*R*)-ECNU-Phos-assisted protocol,[8] were successfully produced with a very high enantioselectivity. In addition, the reaction was accelerated in the case of using (*S*)-**1c**-(*S*)-**1g**, (*S*)-**1i** and (*S*)-**1l** as the substrate, and the corresponding 2,3-allenoates were afforded with a higher enantiopurity in a much shorter time (Table 4, entries 4–8, 10 and 15, as compared with the (*S*)-Segphos-assisted protocol).

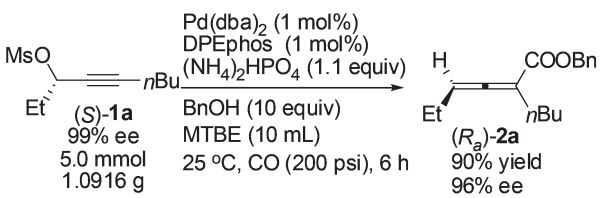
Table 4 Substrate scope of DPEphos-assisted carbonylation of enantioenriched (*S*)-**1a**



	(*S*)-**1**			(R_a)-**2**	
Entry	R^1	R^2	*T* (h)	Yield%[a]	ee%[b]
1	Et	*n*-Bu (**1a**)	9	86 (**2a**)	98
2[c]	Et	*n*-Bu (**1a**)	9	77[d] (**2a**)	97
3	Et	Me (**1b**)	9	69[e] (**2b**)	97
4	Et	*n*-Pr (**1c**)	3	85[f] (**2c**)	97
5	Et	*n*-C_6H_{13} (**1d**)	3	84 (**2d**)	98
6	Et	*t*-Bu (**1e**)	6	84 (**2e**)	99
7	Et	$(CH_2)_2CH(CH_3)_2$ (**1f**)	9	81 (**2f**)	96
8	Et	$(CH_2)_4Cl$ (**1g**)	6	82 (**2g**)	97
9	Et	$(CH_2)_2Ph$ (**1h**)	9	76 (**2h**)	98
10	Et	Allyl (**1i**)	6	82 (**2i**)	98
11	Me	*n*-C_7H_{15} (**1j**)	9	80 (**2j**)	98
12[c]	Me	*n*-C_7H_{15} (**1j**)	9	82 (**2j**)	97
13	*n*-C_7H_{15}	*n*-Bu (**1k**)	9	84 (**2k**)	96
14[c]	*n*-C_7H_{15}	*n*-Bu (**1k**)	9	83 (**2k**)	96
15	i-Bu	*n*-Bu (**1l**)	6	81[g] (**2l**)	97
16[c]	i-Bu	*n*-Bu (**1l**)	6	84 (**2l**)	98

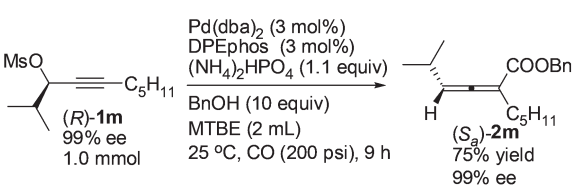
[a] Isolated yield. [b] ee value of the product was determined by HPLC. [c] The reaction was carried out in the presence of 100 psi CO. [d] 97% purity. [e] 92% purity. [f] 98% purity. [g] 95% purity.

Gram scale synthesis of optically active 2,3-allenoate (R_a)-**2a** was also carried out: after 6 h, (R_a)-**2a** was obtained with 90% yield and 96% ee using just 1 mol% $Pd(dba)_2$ and DPEphos (eqn (3)).



(3)

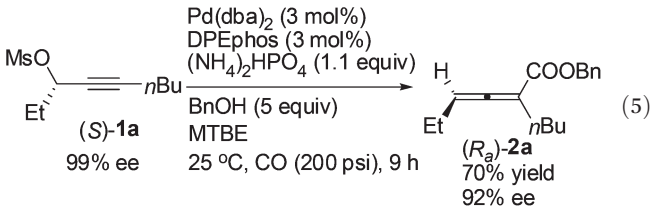
Optically active 2,3-allenoate (S_a)-**2m** was also prepared from the enantiomer (*R*)-**1m**[13] in 75% yield with 99% ee, further affirming that it is a chirality transfer process with the absolute configuration of the 2,3-allenoate being determined by that of the starting propargylic mesylate and the racemization was mostly inhibited in the presence of DPEphos (eqn (4)).



(4)

Reducing the loading of BnOH to 5 equiv. led to a poor result: (R_a)-**2a** was only afforded in 70% yield with a much lower ee (92%) (compare eqn (5) with entry 1 of Table 4).

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(5)

The effect of CO pressure has also been explored.[14] As shown in Fig. 1, the ee value of (R_a)-**2a** increased with the CO pressure in the range of 15–100 psi. And we finally found that the reaction proceeded with the same chirality transfer efficiency even in the presence of 100 psi of CO (97% ee). Some optically active 2,3-allenoates were afforded under these conditions with almost the same results as in the presence of 200 psi of CO (Table 4, entries 2, 12, 14 and 16, compared with Table 4, entries 1, 11, 13 and 15).

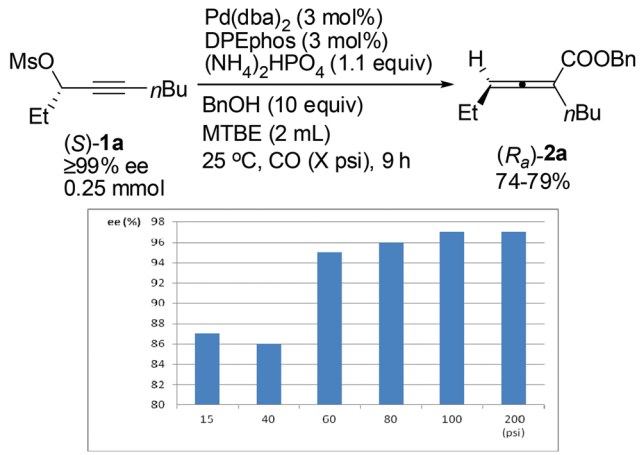
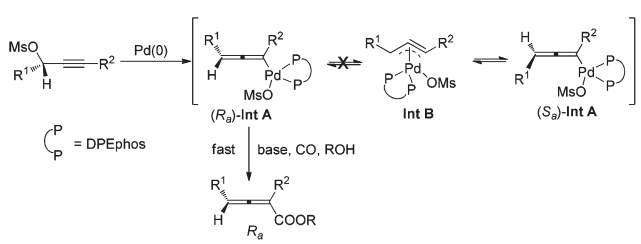


Fig. 1 Effect of CO pressure of DPEphos-assisted carbonylation of enantioenriched (*S*)-**1a**.

Based on these studies, a rationale for the high chirality transfer efficiency in this reaction has been proposed: (1) DPEphos makes the square planar complex (R_a)-**Int A** formed *via* the highly stereoselective S_N2'-type oxidative addition of Pd with the optically active substrates stable enough, avoiding the racemization *via* η^3-type **Int B**, which accounts for the low efficiency chirality transfer efficiency. (2) 10 equiv. of BnOH and a certain level of CO pressure are responsible for the subsequent smooth carboxylation as soon as (R_a)-**Int A** is generated (Scheme 2).[14]



Scheme 2 Rationale for the high chirality transfer efficiency with DPEphos.

In summary, a highly effective protocol for the enantioselective synthesis of optically active 2,4-disubstituted 2,3-allenoates has been established using the inexpensive commercially available DPEphos in the presence of 100–200 psi of CO by starting from the readily available optically active propargylic alcohols. The reaction shows a very high chirality transfer efficiency, providing a very important supplement for the carbonylation of racemic propargylic carbonates using the Pd catalyst and (*R*)-**ECNU-Phos** for 4-non-methyl-alkyl-substituted 2,3-allenoates with ≥96% ee. Through control experiments, it has been unveiled that the matched coordination of Pd with DPEphos and a very fast carboxylation with BnOH and CO are the key for this highly efficient chirality transfer process by preventing racemization. Further studies in this area are being actively conducted in this laboratory.

Experimental

To a flame-dried Schlenk tube were added $Pd(dba)_2$ (28.8 mg, 0.05 mmol) and DPEphos (27.9 mg, 0.05 mmol). Then the tube was degassed and refilled with Ar three times to ensure the complete exclusion of air. Freshly distilled MTBE (2 mL) was subsequently added under argon. The resulting Pd(0) mixture was stirred for 1 hour at room temperature.

To a flame-dried tube filled with Ar were added $(NH_4)_2HPO_4$ (0.7260 g, 5.5 mmol), (*S*)-**1a** (1.1021 g, 5 mmol)/MTBE (4 mL), and BnOH (5.4088 g, 50 mmol)/MTBE (4 mL). After the Pd(0) mixture prepared above was transferred by a syringe to the tube, it was placed in the Parr reactor. The air in the reactor was replaced by CO gas three times sequentially, and the Parr reactor was charged to 200 psi with CO gas. The mixture was stirred at room temperature till the reaction ended, and then the gas was vented in the hood safely. The resulting mixture was diluted with Et_2O (20 mL), washed with brine (20 mL), and dried over Na_2SO_4. After the filtration and concentration under reduced pressure, the crude product was purified by flash chromatography on silica gel to afford (R_a)-**2a** (1.1750 g, 90%) as an oil [eluent: petroleum ether (b.p. 60–90 °C)–ethyl ether = 250/1]: 96% ee (HPLC conditions: Regis (*S*,*S*) Whelk-O column, hexane–i-PrOH = 200/1, 1.0 mL min^{-1}, λ = 214 nm, t_R (minor) = 11.3 min, t_R (major) = 13.0 min); $[\alpha]_D^{21}$ = −37.8 (*c* = 1.21, $CHCl_3$) [96% ee, $[\alpha]_D^{24}$ = −38.0 (*c* = 1.17, $CHCl_3$)]; ^1H NMR (300 MHz, $CDCl_3$): δ = 7.39–7.23 (m, 5 H, Ar–H), 5.63–5.54 (m, 1 H, =CH), 5.22 (d, *J* = 12.6 Hz, 1 H, one proton from Bn), 5.14 (d, *J* = 12.6 Hz, 1 H, one proton from Bn), 2.32–2.18 (m, 2 H, CH_2), 2.18–2.06 (m, 2 H, CH_2), 1.49–1.23 (m, 4 H, 2 × CH_2), 1.04 (t, *J* = 7.4 Hz, 3 H, CH_3), 0.90 (t, *J* = 6.9 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 209.8, 167.6, 136.5, 128.4, 127.8, 127.5, 101.1, 96.7, 66.1, 30.3, 28.1, 22.2, 21.4, 13.9, 13.2.

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11 When 1.5 mol% of $[PdCl(\pi\text{-allyl})]_2$ was used instead of $Pd(dba)_2$ as the catalyst, (R_a)-**2a** was afforded in 78% yield with 94% ee.

12 Recent price per gram according to J&K Chemicals: DPEphos *vs.* 2,2′-bis(dicyclohexyl-phosphino)-1,1′-biphenyl (**L3**) = 80:1890 (in CNY).

13 (*R*)-**1m** was synthesized from the corresponding optically active alcohol, which was obtained *via* kinetic resolution from the racemic one: (*a*) D. Xu, Z. Li and S. Ma, *Chem. – Eur. J.*, 2002, **8**, 5012; (*b*) D. Xu, Z. Li and S. Ma, *Tetrahedron Lett.*, 2003, **44**, 6343. The absolute configuration was affirmed according to the literature: C. R. Reddy, E. Jithender and K. R. Prasad, *J. Org. Chem.*, 2013, **78**, 4251.

14 For details, see Table S1 in ESI.†